

CHAPTER 12

QUANTIFYING UNCERTAINTY

function DT-AGENT(*percept*) **returns** an *action*

persistent: *belief_state*, probabilistic beliefs about the current state of the world
action, the agent's action

update *belief_state* based on *action* and *percept*

calculate outcome probabilities for actions,

given action descriptions and current *belief_state*

select *action* with highest expected utility

given probabilities of outcomes and utility information

return *action*

Figure 12.1 A decision-theoretic agent that selects rational actions.

Proposition	Agent 1's belief	Agent 2 bets	Agent 1 bets	Agent 1 payoffs for each outcome			
				a, b	$a, \neg b$	$\neg a, b$	$\neg a, \neg b$
a	0.4	\$4 on a	\$6 on $\neg a$	-\$6	-\$6	\$4	\$4
b	0.3	\$3 on b	\$7 on $\neg b$	-\$7	\$3	-\$7	\$3
$a \vee b$	0.8	\$2 on $\neg(a \vee b)$	\$8 on $a \vee b$	\$2	\$2	\$2	-\$8
				-\$11	-\$1	-\$1	-\$1

Figure 12.2 Because Agent 1 has inconsistent beliefs, Agent 2 is able to devise a set of three bets that guarantees a loss for Agent 1, no matter what the outcome of a and b .

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Figure 12.3 A full joint distribution for the *Toothache*, *Cavity*, *Catch* world.

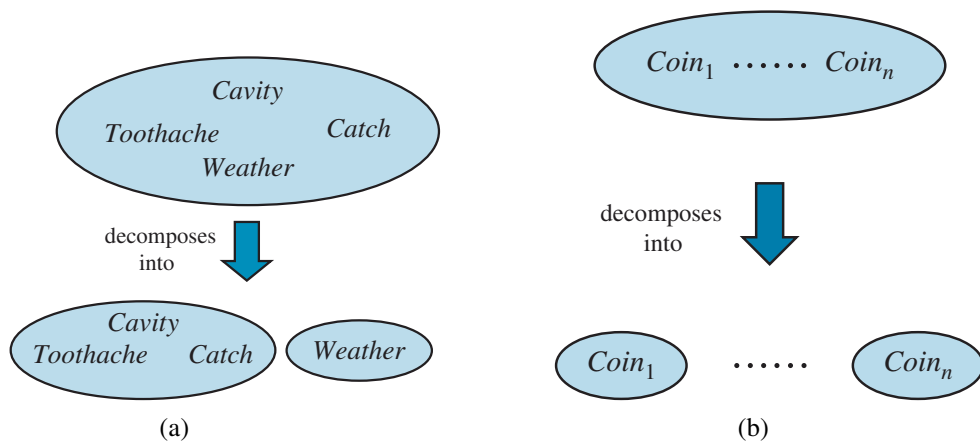


Figure 12.4 Two examples of factoring a large joint distribution into smaller distributions, using absolute independence. (a) Weather and dental problems are independent. (b) Coin flips are independent.

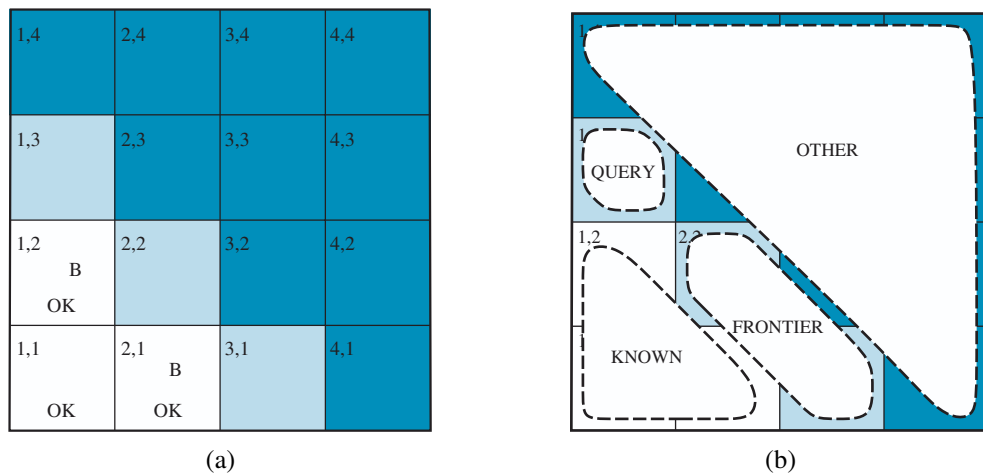


Figure 12.5 (a) After finding a breeze in both [1,2] and [2,1], the agent is stuck—there is no safe place to explore. (b) Division of the squares into *Known*, *Frontier*, and *Other*, for a query about [1,3].

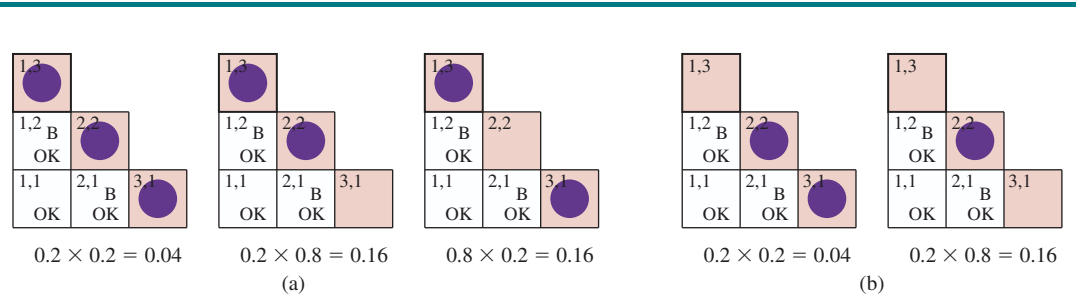


Figure 12.6 Consistent models for the frontier variables, $P_{2,2}$ and $P_{3,1}$, showing $P(\text{frontier})$ for each model: (a) three models with $P_{1,3} = \text{true}$ showing two or three pits, and (b) two models with $P_{1,3} = \text{false}$ showing one or two pits.
